SURVIVING THE FUTURE

Crisis, Climate, and Human Resilience

A Data-Driven, Academically Grounded Guide to Living Through the Crises Caused by Global Warming

Setyo Wisodo

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PREFACE: WHY FUTURE GLOBAL WARMING IS CERTAIN

Global warming is not a hypothesis; it is an observed, measured, and physically explained reality. The certainty of future warming rests on three pillars: (1) the physics of the greenhouse effect, (2) the observed accumulation of greenhouse gases in the atmosphere, and (3) the inertia of the climate system. Even if all emissions stopped today, committed warming from past emissions would continue for decades. This book is written to be debated in academic spaces: every major claim is tied to peer-reviewed literature, data, and expert statements.

The Three Pillars of Certainty

- **Physics:** Greenhouse gases absorb outgoing infrared radiation. This is established physics, tested for over a century (Arrhenius, 1896; modern radiative transfer models).
- **Observation:** Atmospheric CO2 has risen from ~280 ppm (pre-industrial) to ~420 ppm (2024). Global mean temperature has risen ~1.1-1.2 C. These are direct measurements.
- **Inertia:** The ocean stores heat; ice sheets respond slowly. Even with zero emissions, warming continues for decades (committed warming).

Data underpinning the discussion is summarized as follows:

Table P.1 Key global climate indicators (2024)

Indicator	Value	Source
Global mean temperature rise since 1850-1900	~1.1-1.2 C	IPCC (2023)
CO2 concentration (2024)	~420 ppm	NOAA (2024)
Sea level rise since 1900	~20-25 cm	IPCC (2023)
Arctic sea ice decline	~13% per decade	NASA
People affected by climate disasters (annual)	~100 million+	IFRC
Economic damage from climate disasters (annual)	~$200-300 billion	World Bank

The expert consensus underpinning this book is summarized as follows.

James Hansen, former director of NASA’s Goddard Institute for Space Studies, argued in 2013 that two degrees of warming would be enough to expect several metres of sea-level rise, and called the consequences disastrous.

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Katharine Hayhoe, Texas Tech climate scientist (2014, after the IPCC fifth assessment): climate change is here and now rather than in some distant time or place, and the choices made today will shape what the future holds.

Michael Mann, Penn State climatologist (2004; 2021): the scientific consensus on human-caused climate change is comparable to the consensus on gravity, because multiple independent lines of evidence converge.

Friederike Otto, Imperial College and co-founder of World Weather Attribution (2021): the Pacific Northwest heatwave of 2021 would have been virtually impossible without human-caused climate change.

Johan Rockström, Potsdam Institute for Climate Impact Research and Stockholm Resilience Centre (2009 Nature; 2023 update): planetary boundaries define a safe operating space for humanity, and six of the nine are now transgressed.

How This Book Is Organised

The book is arranged so that a reader can enter at three different depths. The first is the argument, which runs through the chapter openings and the sections numbered x.1 to x.3: why the physical certainty of warming is settled, how a local hazard becomes a systemic failure, and what the record of the past two decades shows about which places absorbed an event and which did not. A reader who only has an evening can read those sections across all twenty-four chapters and come away with the claim of the book intact.

The second depth is the case record. Every chapter carries a numbered case study that reconstructs one event in detail — the rainfall totals, the warnings issued, the hours in which power failed, the decisions taken by named institutions — and the analysis that follows the reconstruction. The cases were chosen so that no single explanation covers them: some failed through exposure, some through vulnerability, some through the failure of a warning to reach the people it described, and a few succeeded for reasons that had nothing to do with luck.

The third depth is the appraisal. Each chapter closes with a Future Scenarios section that separates what the physical system has already committed to from what depends on decisions not yet taken, and states plainly which of its own predictions would prove it wrong. That section is the reason the book is meant to be argued with rather than simply read: a prediction that cannot be falsified is a description of the author’s preferences, and the scenarios here are written to be checkable against the record as it accumulates.

Three conventions run through the text. Citations are author and year in the prose, with the full entry in the reference list; figures are the most recent authoritative values available at the time of writing, and where sources disagree the disagreement is stated rather than averaged away. Numbers that describe deaths, displacement or damage are given as the range the sources support, because the false precision of a single figure would misrepresent what is known. And the book avoids the passive construction that hides who decided what: where an institution chose not to act, the text says which

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institution, because responsibility that cannot be assigned cannot be examined.

A note on what this book is not. It is not a manual of climate science, though it summarises the science that the argument depends on. It is not a policy programme, though it ends with the measures that the evidence supports. It is a record of what has already happened, read closely enough to show why the next disaster will fall on some places and not others, and written in the belief that the difference is usually a decision rather than a fate.

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CHAPTER 1

INTRODUCTION: THE WARMING WORLD

Theme: the global crisis of warming and why survival begins with understanding it

In the summer of 2022, a monsoon system carried moisture from the Arabian Sea across Pakistan for eight consecutive weeks (WMO, 2023). By the end of August, one-third of the country was under water, thirty-three million people had been affected, and the national highway network had been cut in more than a hundred places. The flood was not a single event but a sequence: heat that dried the soil, then rain that could not infiltrate it, then reservoir releases that arrived after the rivers had already filled. Six months later, the same country was negotiating with the International Monetary Fund while its farmers reported that two million hectares of cropland would not be planted that season.

A year later, in 2023, Canada burned. More than eighteen million hectares of forest burned, an area larger than England, and the smoke crossed the Atlantic and turned the skies of New York orange (CIFFC, 2023). In September of the same year, two dams above the Libyan city of Derna failed after a storm that had been forecast days in advance; the water killed more than eleven thousand people in a city that had been warned but could not act. In October 2024, a year's worth of rain fell on the Spanish province of Valencia in a few hours and killed more than two hundred people in towns that had never been mapped as floodplains.

These events are usually described as natural disasters. They are not. Each of them required a physical trigger — rain, wind, heat — but the scale of the loss was determined by conditions that people created: where houses were built, how rivers were managed, whether warnings reached the people who needed them, whether the soil had been left compacted and bare, whether the population was old, poor, or living in informal settlements. The trigger is weather. The disaster is a failure of preparation, and preparation is something human beings can change.

This book is an attempt to describe that failure and the ways out of it, in both directions: backward, into the record of what has already happened, and forward, into the range of futures that the physical system has made possible. It is not a book of prophecy. It is a book about the conditions under which communities continue to function, and the conditions under which they do not.

The five years from 2020 to 2024 were the five warmest ever measured, and 2024 was the first calendar year to average more than 1.5 °C above the 1850–1900 baseline (WMO, 2025). That milestone was expected eventually; what surprised researchers was how quickly the instrumental record left the range of previous decades. Ocean heat content reached a record high, and the rate at which the oceans absorbed

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heat in the past decade was roughly double the rate of the 1990s (Cheng et al., 2024). A warmer ocean stores energy that will be released into the atmosphere for centuries, which is why the numbers in this book describe a commitment as much as an event.

1.1 The Decade That Changed the Baseline

The 2020s began with a global pandemic and ended their first half with the four hottest years ever measured. The World Meteorological Organization confirmed that 2024 was the first calendar year to average more than 1.5 °C above the 1850–1900 baseline, at about 1.55 °C, with 2023 close behind at 1.45 °C (WMO, 2025). The two years were lifted by a strong El Niño, but the underlying trend was not: the previous ten years were the ten warmest in the instrumental record, and every one of them exceeded the average of the decade before it.

The warming has not been evenly distributed in time or space. The Arctic has warmed three to four times faster than the global mean; the Mediterranean and the Middle East have warmed roughly twice as fast; the North Atlantic has cooled in some patches even as the ocean as a whole has gained heat, because melting ice and shifting winds have altered circulation. A global average of 1.5 °C therefore describes a world in which some regions have already experienced 2.5 °C or 3 °C of local warming, and in which the statistics of extreme heat have shifted far more than the mean (IPCC, 2021).

The decade also changed the character of the evidence. Attribution studies that once took a year now take a month. Satellite records that began in the 1970s now span half a century. Insurance losses, crop models, hospital admissions and migration data have accumulated to the point where the effect of climate on society can be measured rather than inferred. The result is a body of evidence that is no longer primarily about the future: it is about what has already happened, to identifiable people, in identifiable places, at a cost that can be counted.

The reason a single year matters is not the year itself but the shift in the distribution that it reveals. A global average is built from billions of hourly readings, and a rise of 0.1 °C in that average corresponds to a large increase in the number of days that cross a threshold somewhere — the temperature at which a road softens, a power line sags, an outdoor worker stops being able to cool, or a night stops providing recovery from the day before (IPCC, 2021). Public discussion treats warming as a slow trend; in practice it is experienced as a change in the frequency of extremes, which is why the same average rise produces very different consequences in different places.

The phrase ‘new normal’ is therefore misleading. There is no stable baseline to which planning can return: every decade is warmer than the last, and the statistics that engineers, insurers and farmers rely on describe a climate that has already gone. That is why the chapters that follow use moving baselines rather than long historical averages, and why a design that was adequate in 1990 can fail in 2030 without any component of it having deteriorated.

The arithmetic of a baseline deserves a moment. Climate normals are recalculated every decade, which means the reference against which a day is called hot moves with the climate itself. A summer that would have looked extreme against the

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1961–1990 average sits close to ordinary against the 1991–2020 average, and the change happens without any single event announcing it. Farmers register the shift in planting dates, insurers in the frequency of claims, and municipal engineers in the depth of storm drain they must install. The result is a quiet form of forgetting. Each generation inherits the weather of its childhood as normal and measures change against a baseline that has already moved, which is why older residents and younger ones can argue about the same summer and both be describing what they actually experienced.

1.2 The Anatomy of a Global Crisis

A global crisis is not simply a large disaster. It is a set of stresses that arrive together, reinforce one another, and exceed the capacity of institutions to respond. Climate change qualifies because it acts on every sector at once: it raises food prices by reducing yields, raises insurance costs by increasing losses, raises migration by making some places uninhabitable, and raises the cost of borrowing by increasing uncertainty. Each effect is modest in isolation. Together they consume the fiscal and administrative space that governments need to cope with the next shock.

The reinforcement is the part that is usually missing from public discussion. A drought reduces the harvest; the reduced harvest raises food prices; higher food prices increase poverty; poverty increases vulnerability to the next flood; the flood destroys the assets that the household would have used to recover. The sequence can turn a single bad season into a decade of lost development, and it explains why the same hazard produces a few dozen deaths in one country and thousands in another.

This is also why the language of ‘natural disaster’ is misleading in a second sense. It suggests a boundary between the hazard and the society exposed to it, when in reality the boundary is the thing that policy controls. A levee, a building code, a crop insurance scheme, a heat action plan and a functioning weather service are all ways of moving that boundary. The chapters that follow examine each of them, but the logic is the same throughout: the physical hazard sets the pressure, and the social system determines whether the pressure becomes a catastrophe.

The scale of the crisis is best measured in the language of risk rather than catastrophe: hazard multiplied by exposure multiplied by vulnerability. The United Nations Office for Disaster Risk Reduction estimates that the number of recorded disaster events has roughly quadrupled since the 1970s, driven less by more storms than by more people and assets in harm’s way (UNDRR, 2022). Average annual losses from disasters are now in the hundreds of billions of dollars, and the losses fall hardest on countries with thin insurance markets and limited fiscal space, so that a single event can erase years of development spending.

The timing matters as much as the overlap. Crises become unmanageable when the interval between them is shorter than the time needed to recover, and that interval has been shrinking in many regions. A flood that arrives before the previous flood’s repairs are finished leaves the same households in temporary shelter twice, exhausts the same municipal budget, and teaches people that rebuilding is pointless. Recovery time is not a fixed quantity either: it depends on insurance payments, labour, materials and

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permits, all of which are themselves disrupted by the event. The practical consequence is that a sequence of moderate hazards can do more cumulative damage than a single severe one, and that disaster statistics built on individual events systematically miss this.

A hazard becomes a global crisis through four channels, and each one transmits harm further than the event itself. Trade is the first: a harvest failure raises the price of a staple in every market that imports it, and a factory that loses power delays components for assembly lines on other continents. Finance is the second: a sovereign that borrows to rebuild raises its debt service, and the cost is paid in the budget for schools and clinics. Movement is the third: people displaced from a coast or a delta arrive in cities that were not prepared for them. Information is the fourth, and it works in both directions, because a rumour travels faster than a warning and a photograph can mobilise help or panic.

1.3 From Local Hazard to Systemic Failure

Most climate impacts begin as local events, but their consequences travel through markets, supply chains and migration routes. The 2011 floods in Thailand are the standard example: a sequence of monsoon rains inundated industrial estates north of Bangkok, global hard-drive production fell by about a quarter, and computer prices rose around the world. The flood was national; the shortage was global. The same mechanism now operates through semiconductor production in Taiwan, wheat exports from the Black Sea, and container shipping through the Red Sea.

Systems fail when a shock exceeds redundancy. A power grid with spare capacity survives a heatwave; a grid without it cascades. A port with alternative routes survives a blockage; a port that is the only route does not. The 2021 Texas winter storm, the 2022 European gas crisis, and the 2023 Panama Canal drought all demonstrated the same principle in different sectors: the failure was not caused by the weather alone but by the absence of slack that would have absorbed it. Climate change is, among other things, a systematic reduction in slack.

For households, the translation is direct. A family that depends on one income, one water source, one road and one food market is exposed to every shock that touches any of them. Diversification — of income, of water storage, of food sources, of social contacts — is the household equivalent of redundancy in a grid. It is also the reason preparedness is not a single purchase but a set of overlapping capabilities, each of which buys time when another fails.

The systemic dimension is what makes climate risk different from a sequence of bad accidents. Modern societies depend on tightly coupled networks — power, water, transport, communications, finance, health — and a shock that begins in one network can travel through the others faster than any single agency can respond. The Texas winter storm of 2021 is the clearest recent example: a fuel-supply failure became a power failure, which became a water failure, which became a health emergency. Analysts describe this as cascading risk, and it means that the vulnerability of a system is set by its weakest coupling, not by its strongest component.

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River transport is the clearest recent example. In 2018 and again in 2022, low water on the Rhine forced barges to run partly loaded, and the river carries a large share of Germany's coal, chemicals and mineral oil. Chemical plants that depend on continuous barge deliveries cut output; the price of transport rose; and the effects reached factories that had no direct contact with the river. Nothing in that chain was unusual in isolation — low water happens, and supply chains reroute — but the combination of a slow-onset hazard and a tightly scheduled production system turned a hydrological fact into an industrial one. The same pattern appears wherever a system has removed its own buffers: a just-in-time warehouse, a single-source supplier or a single bridge can transmit a local shock across a continent.

The mechanism that turns a local hazard into a systemic failure is dependency rather than severity. A storm that closes a port interrupts the fuel supply to the generators that keep a hospital running; a flood that submerges a substation disables the pumps that would have drained the streets; a heatwave that melts a road prevents the trucks that carry the water. Each link is individually ordinary and collectively fragile, which is why the loss from a moderate event in a tightly coupled system can exceed the loss from a severe event in a loosely coupled one. Engineers call this a cascade, insurers call it a contingent business interruption, and the household that loses power, water and wages at the same time experiences it as a single catastrophe.

1.4 The Human Cost: What the Record Shows

The human cost of climate change is usually reported through disasters, and the disaster statistics are stark enough. The Centre for Research on the Epidemiology of Disasters recorded more than 7,000 major disasters between 2000 and 2019, affecting 4.2 billion people and killing 1.2 million (UNDRR, 2022). Weather and climate events accounted for the large majority of them. In the 2020s the pattern has continued: the 2022 Pakistan floods affected thirty-three million people; the 2023 Turkey–Syria earthquakes — geological rather than climatic, but a reminder of compound vulnerability — killed more than fifty thousand; the 2024 Valencia floods killed more than two hundred in a high-income country with a modern warning system.

But disasters capture only the visible part of the cost. The World Health Organization projects that climate change will cause about 250,000 additional deaths a year between 2030 and 2050 through heat, diarrhoea, malaria and undernutrition, and the true toll is probably higher. Heat is already lethal: the European summer of 2022 is estimated to have caused about 61,700 deaths, most of them among older people (Ballester et al., 2023). Air pollution, much of it from the same fuels that drive warming, kills about seven million people a year.

The economic record is equally uneven. The World Bank estimates that climate-related losses in developing countries already exceed US$200 billion a year, and that the figure understates the true cost because it counts damaged assets rather than lost development (World Bank, 2021). A child who is malnourished during a drought does not recover the cognitive development that was lost; a farmer who sells livestock to survive a season does not rebuild the herd at the same rate. These effects are rarely

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counted in disaster statistics, but they are the mechanism by which a temporary shock becomes a permanent reduction in welfare.

The human cost is systematically undercounted, and the undercount is not accidental. Deaths from heat are recorded as cardiovascular or renal failure; deaths from drought are recorded as malnutrition or infection; deaths from displacement are recorded in the places people move to, often months later. The most careful estimates come from excess mortality analysis, which compares observed deaths with the number expected in a normal period. Those studies consistently find tolls several times larger than official counts, and they are the reason this book prefers ranges to single figures.

What the record misses is as revealing as what it contains. Disaster databases count deaths, injuries and damaged buildings, because those are the categories that agencies collect. They do not count the school year lost when a building is used as a shelter, the vaccination schedule interrupted when clinics close, the harvest that is smaller because the soil was washed away, or the year of income lost by a family that moved to a city and found informal work. These losses are real, cumulative and largely invisible in the statistics that governments use to set budgets. The consequence is a systematic bias in public reasoning: hazards that kill quickly and visibly attract money, while hazards that erode health, learning and income over years are treated as background conditions.

The human cost of a decade of extremes is easiest to grasp through a handful of numbers and hardest to grasp through any of them. More than half a million people have died in the heat events of the last quarter-century; floods have displaced tens of millions in a single season in a single country; a drought in one region has killed a quarter of a million people before a famine was declared (Romanello et al., 2024). Each number is an estimate with a method behind it, and the method matters because it determines who is counted. The deaths that are recorded are the ones that a system noticed, and the people who are most exposed are often the ones whose deaths are least likely to be recorded as anything in particular.

1.5 Why the Worst Outcomes Are Not Yet Fixed

The physical system has inertia, and that inertia is usually presented as bad news: the warming already committed will continue for decades, sea level will rise for centuries, and some changes are irreversible. All of that is true. But the same inertia means that the future is not yet determined, because the emissions that would produce the worst outcomes have not yet happened. The difference between 1.8 °C and 4.4 °C at the end of the century is not a difference in physics; it is a difference in cumulative carbon, and cumulative carbon is still being decided (IPCC, 2021).

The remaining carbon budget makes the arithmetic explicit. To have a fifty per cent chance of holding warming to 1.5 °C, the world can emit roughly 250 billion tonnes of carbon dioxide more — about six years at current rates (IPCC, 2021). For 2 °C the budget is roughly 1,150 billion tonnes. These numbers are uncertain at the margins, but their order of magnitude is not, and it implies that every year of delay consumes a substantial share of the remaining budget. It also implies that the choice is not

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between action and no action but between action now and much larger, more expensive action later.

There is a second reason for qualified optimism. The cost of the technologies that reduce emissions has fallen faster than almost anyone predicted. Solar photovoltaic costs have fallen by roughly ninety per cent since 2010, onshore wind by about seventy per cent, and lithium-ion battery packs by about eighty-five per cent (IRENA, 2023). In 2023 the world added more renewable generating capacity than fossil capacity for the first time, and the International Energy Agency now expects renewables to account for the majority of new power capacity through 2030. The obstacle is no longer primarily engineering; it is the speed at which grids, permits and political coalitions can change.

Some of the worst outcomes are already locked in by physics, and some are not. The distinction matters for policy. Warming that has already occurred commits the world to further sea-level rise and to more intense heat extremes, because the ocean and ice sheets respond slowly to emissions. Other outcomes depend on decisions still being taken: how quickly power systems are rebuilt, whether cities invest in drainage, whether social protection reaches the households that need it. The book is organised around that distinction between the committed and the contingent.

It helps to separate three kinds of lock-in, because they respond to different policies. Physical lock-in is the warming already stored in the ocean and the sea-level rise already committed by ice that has begun to move; it cannot be reversed on any human timescale. Infrastructure lock-in is the power plants, buildings and roads that will operate for decades and whose emissions are effectively decided when they are built. Behavioural lock-in is the pattern of settlement and consumption that makes high emissions convenient — the suburb without transit, the office without shade. Only the second and third are still open to rapid change, and they are where the leverage lies. A tenth of a degree avoided is not a rounding error; it is a smaller tail of extremes, a shorter fire season and a slower rate of sea-level rise for everyone who comes after.

The worst outcomes are not yet fixed because the fix is institutional rather than technological, and institutions move at the speed of politics. The warning system that would have saved the people who died in a flood is usually available and usually unfunded; the building code that would have kept a roof on is on the books and not enforced; the evacuation plan that would have emptied a valley was written and never rehearsed. None of that is a mystery, and none of it is expensive in the terms of a national budget. What it requires is the patience to fund maintenance, the courage to enforce a code against a developer, and the habit of treating a forecast as an instruction rather than as information.

1.6 What ‘Surviving’ Actually Means

The word survival suggests a single moment: the storm passes, the flood recedes, the family is alive. In practice, survival is a process that extends long after the water leaves. A household that survives the flood but loses its documents, its savings and its access to credit may not survive the next two years. A city that survives the heat-wave but loses its power grid for a week may lose thousands of people to hospitals that

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cannot run. The relevant question is not whether a system survives the event but whether it returns to a level of functioning from which it can absorb the next one.

This book therefore treats survival at three levels. Personal survival is about water, food, shelter, health and the knowledge to use them. Community survival is about mutual aid, local institutions and the physical infrastructure that connects households to services. Societal survival is about the fiscal capacity, the legitimacy and the administrative competence that allow a state to keep functioning when several crises arrive at once. The levels are nested: a household cannot be secure in a failed state, and a state cannot be stable when its households are desperate.

The nesting also explains why purely individual preparation is insufficient. A family that stores water and food is better off than one that does not, but it still depends on a functioning water utility, a supply chain, a hospital and a road. Preparedness that ignores the collective is a bet that everyone else will continue to provide the services on which the prepared household depends. That bet may pay off in a short emergency; it fails in a prolonged crisis, which is precisely the scenario that climate change makes more likely.

Survival, as this book uses the term, has three thresholds rather than one. The first is immediate physical safety: staying out of floodwater, avoiding heatstroke, having air to breathe during a smoke episode. The second is recovery: retaining shelter, income, documents and health care long enough to rebuild. The third is systemic: living in a place whose institutions can absorb the next shock without collapsing into scarcity and conflict. Most public discussion focuses on the first and neglects the second and third, which is why so many people survive the event and are ruined by the aftermath.

Two words are often confused in this discussion. Resilience is the capacity to absorb a shock and return to the previous state; robustness is the capacity to keep functioning without returning to it. A city that rebuilds the same bridge in the same place after every flood is resilient in the first sense and foolish in the second. The distinction matters because climate change moves the baseline: returning to the previous state means returning to a state that is already too exposed. The more useful goal is to bend without breaking and to change deliberately rather than under pressure — to move the hospital generator above the new flood line before the flood, not after it, and to widen a road for evacuation while the weather is quiet.

Surviving, in the sense this book uses the word, is not the same as being rescued. Rescue is an event that happens to a person; survival is a condition that a household, a community and an institution maintain over time, through the event and the months that follow. A household that is evacuated and then spends two years in a container is not surviving in this sense, and neither is a city that rebuilds its waterfront to the old standard. The test is whether the arrangements that protected people in one event are still in place for the next one, and whether they protect the people who were least able to protect themselves.

It is worth turning the definition into four tests that a household, a city or a country can apply to itself without a consultant. The first is continuity: would the water, the power, the clinic and the school keep working, at a reduced level, for a week. The second is reach: would the warning and the assistance arrive at the household that is hard-

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est to reach, which is usually the one headed by an older woman in a settlement without a street address. The third is recovery: would the household be back in a habitable home and an income within a year, or would it be in a container for three. The fourth is memory: would the institution that failed be able to say what it changed, and would the change still be funded after the next election. A place that answers yes to all four is surviving in the sense this book means, and very few places answer yes to more than two.

1.7 Knowledge, Preparedness and Adaptive Capacity

Adaptive capacity is the ability of a system to adjust to change while remaining functional. It is not the same as wealth, though wealth helps; it is the combination of information, resources, institutions and social trust that allows a community to anticipate, absorb and recover. Bangladesh has far less wealth than the United States, but its cyclone preparedness is widely regarded as superior, because it has invested in shelters, volunteers, early warnings and drills that reach the village level. Adaptive capacity is therefore a choice about where to invest, not simply a by-product of income.

Knowledge is the cheapest component and the one most often neglected. A household that knows which road floods first, which neighbour has a generator, and which clinic stays open can make better decisions in the first six hours than one that has to discover those facts in the middle of the emergency. A city that knows which neighbourhoods have the oldest residents and the least tree cover can target cooling measures before the heatwave arrives. The information is rarely secret; it is simply not assembled, not practised, and not shared.

The chapters that follow treat knowledge as infrastructure. That means writing down the plan, testing it, updating it, and making sure the people who need it can find it in a language and format they can use. It also means distinguishing between knowledge that is stable — the physics of heat, the geography of a floodplain — and knowledge that changes, such as the location of a shelter or the phone number of a clinic. A plan that does not distinguish between the two will either become obsolete or become too vague to act on.

Adaptive capacity is the product of resources, institutions, information and social connection, and its determinants have been studied for decades. Yohe and Tol (2002) showed that the capacity to adapt depends on economic wealth, technology, education, information and social capital, and that these factors interact rather than substitute for one another. A wealthy city without trusted information adapts poorly; a poor community with strong networks and accurate warnings often adapts better than its income would predict. That finding is the reason this book gives as much attention to institutions and relationships as to technology.

Adaptive capacity is usually described as a stock of four capitals: human (skills, health, education), social (networks, trust, institutions), physical (infrastructure, savings, equipment) and natural (soil, water, ecosystems). Households and cities differ in the mix, and the mix explains why income alone predicts survival poorly. A poor neighbourhood with dense kinship networks and a functioning mosque, church or community association often responds faster than a richer one where nobody knows their

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neighbours. The last mile of any warning system runs through those networks. A forecast that reaches a phone but not a person who can act, in a language that person understands, at an hour when acting is possible, is not a warning; it is data.

Adaptive capacity has three components, and they do not advance at the same speed. The first is economic: a household with savings, insurance and a second income can absorb a shock that would ruin a household without them. The second is technical: a country with a modern meteorological service, a functioning grid and a building code can convert a warning into an action. The third is institutional, and it is the one that lags, because it depends on the unglamorous work of mandates, budgets and accountability. The evidence from four decades of disaster research is that the third component explains most of the variation in outcomes between countries with similar incomes and similar hazards.

1.8 The Argument of This Book

The argument can be stated in four propositions. First, the climate has already changed enough to produce measurable harm, and the harm will increase because of warming that is already committed. Second, the harm is not distributed evenly: it falls hardest on people with the fewest resources, in places with the weakest institutions, and on hazards — heat, drought, flood — that are least insured. Third, the capacity to reduce harm exists and is largely a matter of preparation, governance and investment rather than of inventing new technology. Fourth, the difference between a manageable future and a catastrophic one is still a political choice, made year by year, in decisions about energy, land use and finance.

The book is organised around those propositions. The first part establishes the physical and observational basis: how warming works, how it is measured, and why the record is trustworthy. The second part examines the main channels through which harm arrives — heat, water, food, ecosystems, health, economies, migration and conflict. The third part turns to response: infrastructure, education, information, household preparedness, emergency health, social capital, technology and governance. The fourth part presents case studies and closes with action plans that a household, a community or a city can use.

Each chapter follows the same discipline. It begins with field conditions — what has actually happened, where, and to whom. It then brings in the research literature to explain why those conditions produced those outcomes, and it distinguishes between what is well established, what is contested, and what is not yet known. It ends with a set of future scenarios built from historical data, present observations and explicit assumptions. The aim is not to persuade the reader of a conclusion but to give the reader the materials to reach one.

The method of the book is deliberate. Each chapter begins with a documented field condition — a storm, a drought, a blackout, a hospital, a harvest — and asks what actually happened and why. It then brings in the research literature that has examined the same question, including where the literature disagrees. It closes by combining the two into an argument about what the case implies for the next decade. Where the evi-

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dence is thin, the text says so. Where an estimate is contested, it gives the range rather than a single number.

The four propositions are arranged so that each one constrains the next. If the harm were not already measurable, adaptation would be speculative; if it were not going to grow, preparation could wait; if it were not unevenly distributed, a single global policy would suffice; and if the worst outcomes were fixed, planning would be pointless. Each chapter therefore tests one link in that chain rather than restating the conclusion. The chapters also refuse a habit common in climate writing: the confident single number. Where the science gives a range, the book gives a range; where the evidence is thin, it says so; and where a projection depends on a policy choice, the choice is named rather than hidden inside the figure.

The argument of this book is that the difference between a hazard and a catastrophe is a set of arrangements, and that the arrangements are known, affordable and mostly boring. It is not an argument that disasters are avoidable, because they are not, nor that technology will rescue the century, because it will not. It is an argument about the margin. Two countries can experience the same storm and record death tolls that differ by two orders of magnitude, and the difference is rarely the storm. Reading the chapters that follow as a single argument means watching for the same variables in each of them: who was warned, who could move, who was insured, who was counted, and who was forgotten.

1.9 How the Chapters Fit Together

The first six chapters are diagnostic. Chapters 1 to 3 establish the physical science and the observation systems; chapters 4 to 6 examine heat, water and flooding as the three hazards that account for most direct losses. Chapters 7 to 12 follow the consequences through food, ecosystems, health, economies, migration and conflict. These chapters are ordered by the sequence in which a shock typically propagates: heat and water arrive first, food and health follow within weeks, economies and migration follow within months, and conflict, where it occurs, emerges from the interaction of all of them.

Chapters 13 to 21 are about response. They cover infrastructure, education, data and misinformation, household preparedness, emergency health, social capital, renewable energy, psychology and governance. The order is deliberate: physical infrastructure is the most visible response and the slowest to build; education and information are the cheapest and the most durable; social capital is the resource that determines whether the other responses reach the people who need them. Governance appears last because it determines whether any of the others can be sustained.

Chapters 22 to 24 are applied. Two chapters of case studies examine what happened in specific places — heatwaves, fires, floods, droughts, food crises and security emergencies — and what distinguished the responses that worked from those that did not. The final chapter provides action plans at three scales: the household, the community and the city. Readers who need immediate guidance can begin there and return to the analytical chapters later; readers who want to understand the reasoning can read in

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order.

The chapters are arranged in three arcs. The first arc, chapters 2 to 6, establishes the physical and human baseline: how the climate system works, how heat and water and ice are changing, and what those changes do to cities and coasts. The second arc, chapters 7 to 13, follows the consequences through food, ecosystems, health, money, migration, conflict and infrastructure. The third arc, chapters 14 to 24, turns to response: literacy, data, household preparation, emergency health, social capital, energy, psychology, justice and action plans. Readers who need a particular answer quickly can use the contents; readers who want the argument should follow the arcs.

The three arcs are also three questions. The diagnostic arc answers what is happening and why; the response arc answers what can be done, by whom and at what cost; the applied arc answers what it looks like when a specific place tries. The order matters, because a reader who begins with the checklists in chapter 24 without the diagnosis in chapters 3 to 12 will prepare for the wrong hazard. The case studies in chapters 22 and 23 sit at the end deliberately: they are not illustrations of the argument but tests of it, and each one is chosen because some part of the story refused to behave as the general model predicted.

The chapters are arranged in four parts, and the arrangement is deliberate rather than chronological. The first part establishes the physical baseline: what is warming, by how much, and how it is measured. The second part follows the harm into the systems that people depend on: food, water, cities, health, energy. The third part examines the institutions that determine whether the harm becomes a disaster: warning, evacuation, insurance, reconstruction, public health. The fourth part looks forward, at the scenarios, the economics, the justice claims and the practical checklist. A reader with limited time can start with the fourth part and work backward, because it names the decisions that the earlier chapters explain.

1.10 Reading in an Emergency

A book is a poor instrument in an emergency, and this one is designed with that limitation in mind. If a hazard is imminent, the priority is to act on the guidance of local authorities, not to consult a chapter. If the hazard has passed and the household is safe, the chapter on household preparedness provides a checklist for the next seventy-two hours and the next three months. If the reader is a planner or a teacher, the chapters on infrastructure, education and governance are the relevant starting points.

The book is also designed to be updated. Climate science, disaster statistics and technology costs change faster than books can be reprinted, and a reader who relies on a number without checking its date will eventually be misled. Each chapter therefore identifies the kind of evidence that would change its conclusions, and the final chapter lists the data sources that are updated most frequently. The habit of checking the date of a claim is itself a survival skill, because misinformation often consists of true statements that were true once.

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Finally, the book is written for readers who are not specialists. Technical terms are explained when they first appear; numbers are given with their units and their dates; and uncertainty is stated rather than concealed. The aim is not to make the reader a climate scientist but to make the reader capable of following an argument, detecting an exaggeration, and deciding what to do. That capacity — not the memorisation of facts — is what allows a community to adapt when the facts change.

A book is not an emergency plan. If a warning has been issued where you are, the authority's instructions take precedence over anything printed here, because they reflect conditions on the ground that no book can know. What this chapter offers instead is preparation for the hours when instructions are late, contradictory or absent: knowing which hazards affect your area, where the nearest shelter is, how much water you have, who depends on you and who will check on you. Those answers are collected in chapter 16 and summarised in chapter 24, and they are worth having before you need them.

The book can be used at three speeds. In the twenty-four hours before a known hazard, chapter 24 and the checklists in chapter 16 are the only pages that matter. In the weeks after, chapters 17 and 18 cover health, shelter, finance and the reconstruction politics that decide whether recovery is fair. In the months between events — the normal condition of most places — the diagnostic and response chapters are the ones worth reading slowly, because that is when a household, a school or a council has the time to change something. The one instruction that overrides the rest is to trust the local authority's warning over any general advice printed here, since the book cannot know the street, the tide or the bridge.

A book read in an emergency is read differently from a book read in a library. The reader who is preparing for a season will want the checklists and the sections on warning and evacuation; the reader who is recovering will want the chapters on insurance, claims and reconstruction; the reader who is teaching will want the case studies and the data. The chapters are written so that each can be entered without the ones before it, and the cross-references point to the section rather than to the page. The one habit worth adopting is to read a chapter and then explain it to somebody else, because the act of explanation is the most reliable test of whether the material has been understood.

1.11 Case Study: The 2022 Pakistan Floods

The floods began in June 2022 with monsoon rain that was heavy but not unprecedented. What made them catastrophic was the sequence. A spring heatwave had melted snow and glaciers early and baked the soil into a crust; when the rain arrived, it ran off instead of infiltrating. The Indus and its tributaries rose together, and the dams and barrages that had been built to control them released water downstream to protect their own structures. By late August, the government had declared a national emergency, and satellite imagery showed a lake stretching across Sindh.

The numbers are a case study in how climate impacts are counted. About 33 million people were affected, 1,700 died, and 2.2 million houses were damaged or destroyed. The Post-Disaster Needs Assessment estimated damage at US$14.9 billion

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and economic losses at US$15.2 billion, together about 4.8 per cent of GDP (UNDRR, 2023). Agriculture lost most of a season: two million hectares of crops were destroyed, and the cotton harvest fell by about a third. The damage was concentrated in Sindh and Balochistan, the two poorest provinces, which is a pattern repeated in disaster after disaster.

Attribution research found that climate change made the extreme rainfall more likely, though the magnitude of the effect varied between studies and depended on the definition of the event (Clarke et al., 2022). But attribution explains only the trigger. The vulnerability had been built over decades: irrigation systems that had silted up, embankments that had not been maintained, housing in floodplains, and a population whose livelihoods depended on a single crop season. The flood was a climate event; the disaster was a development event.

The response revealed both the strengths and the limits of international aid. Donor pledges eventually exceeded US$10 billion, but they arrived over months, while the most urgent needs were in the first weeks. The Pakistani military, which has the country's largest logistics capacity, led the relief effort; provincial governments coordinated shelters; and local charities, particularly religious foundations, distributed food faster than the formal system. The pattern — rapid local response, slow international finance — is typical, and it is why the chapters on governance and social capital treat local institutions as part of the infrastructure.

Recovery has been uneven. By 2024, many households had rebuilt with the same materials in the same places, because they had no alternative and because reconstruction assistance covered only a fraction of the cost. The lesson is not that Pakistan should have prevented the rain. It is that recovery which restores the pre-existing vulnerability is not recovery; it is the preparation of the next disaster. The same lesson appears in the case studies of every chapter that follows.

For readers outside Pakistan, the case has three implications. It shows that a climate hazard can erase years of development in a single season. It shows that vulnerability is produced by decisions about land, water and housing that are made long before the hazard arrives (Adger, 2006). And it shows that the difference between a disaster and a catastrophe is often the speed and reach of the response — which is a matter of institutions, not of meteorology.

The official Post-Disaster Needs Assessment, conducted jointly by the government of Pakistan and international institutions, put total damage and losses at more than US$30 billion and estimated recovery needs at about US$16 billion, of which a fraction had been committed when the pledging conference closed (World Bank, 2022). The gap between assessed need and available finance is the practical form that climate injustice takes: a country with less than one per cent of cumulative emissions financing its own reconstruction while servicing external debt.

The geography explains the sequence. Pakistan's upper Indus basin holds thousands of glaciers, and the spring heat that preceded the monsoon accelerated their melt; the rivers carried that water down to plains that had already lost soil moisture to the same heat, so the rain that followed ran off rather than soaking in. In the south, Lake Manchar rose until engineers deliberately breached a protective embankment in

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September to keep the water away from larger towns, flooding villages that had been told to evacuate and some that had not. The result was a flood that lasted for months in places, not days, and that produced a second wave of harm — waterborne disease, lost winter wheat, and hundreds of thousands of people still in tents when the cold season arrived.

The relief operation that followed the Pakistan floods of 2022 was constrained by a factor that the photographs did not show: the road network (WMO, 2023). The water did not recede for months in the southern provinces, the embankments that would have protected the towns had been breached or built in the wrong place, and the villages that were cut off could be reached only by boat. The humanitarian system responded with a scale that was unprecedented for the country, and the coordination between the government, the United Nations and the local organisations was uneven. The lesson that the evaluation drew was that the response was limited less by the supply of aid than by the accessibility of the people who needed it.

A second flood makes the same point from another continent. In March 2019, Cyclone Idai came ashore near Beira, Mozambique, and pushed the sea up the Pungue and Buzi rivers instead of letting it drain away; more than a thousand people died across the three affected countries and roughly three million were affected, in a city whose drainage had not been maintained for years (IFRC, 2022). Pakistan and Mozambique share the mechanism: extreme rainfall is only half of a flood, and the other half is the condition of the ground the water crosses. Neither country lacked a warning. Both lacked the capacity to act on one, which is why the next two decades cannot be forecast from rainfall alone.

1.12 Future Scenarios and Predictions: What the Next Two Decades Hold

Everything that follows in this book rests on a trend that is already measurable. Since 1970, global surface temperature has risen by about 0.18 °C per decade, sea level by about 3.4 millimetres per year, and the frequency of extreme heat has increased by a factor of roughly three in many regions (IPCC, 2019b). The number of weather-related disasters reported to EM-DAT has risen from about 100 a year in the 1970s to more than 300 a year in the 2010s, though part of that increase reflects better reporting. The losses, adjusted for inflation, have grown by about 5 to 7 per cent a year, faster than exposure alone would predict.

The present position is that the world is on a trajectory consistent with about 2.7 °C of warming by 2100 if current policies are maintained, according to the International Energy Agency’s assessment of announced pledges and implemented measures. That is below the 4.4 °C of the highest scenario but well above the 1.5 °C target (IPCC, 2021). The near-term trajectory is largely locked in by emissions already released and by infrastructure already built: even if every country met its net-zero pledge, the next two decades would still warm by about 0.4 °C above the current level (IPCC, 2021).

The scenarios below are therefore not predictions of what will happen but conditional projections of what could happen under stated assumptions. They differ mainly in the speed of decarbonisation and in the response of slow variables such as ice sheets and

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the overturning circulation. Each is anchored to observed data: the 2024 concentration of 424 parts per million, the 1.55 °C annual anomaly, the 4.5 millimetres per year of sea-level rise, and the current rate of energy transition (NOAA, 2024).

The next two decades can be read as a sequence rather than a blur. The 2020s are the decade in which individual years cross 1.5 °C while the multi-year average remains below it; the 2030s are the decade in which the average itself approaches that line under current policies; and by the 2040s the difference between the high and low paths becomes visible in the frequency of extremes rather than only in the global mean (IPCC, 2021). Four indicators are worth watching because they move early and are hard to fake: ocean heat content, the mass balance of the Greenland and Antarctic ice sheets, the extent of Arctic sea ice in September, and the fraction of the land surface in drought at any moment. Each is measured continuously, published openly and difficult to reinterpret.

Table 1.1 Three futures for the next two decades

Scenario	Assumption	2035	2050	Signature impacts
Accelerated transition	Emissions peak before 2025, halve by 2035, net zero by 2060	~1.7 °C	~1.8 °C	Heat extremes stabilise; sea-level rise ~0.3 m by 2050; coral reefs still decline
Current policies	Implemented measures only, slow tightening after 2030	~1.9 °C	~2.1 °C	Severe heatwaves annual in South Asia; crop losses 5–10%; 200 million displaced by 2050
Delayed transition	Emissions plateau until 2040, rapid catch-up afterwards	~2.1 °C	~2.4 °C	Arctic summers ice-free; reef collapse; 400 million in coastal flood zones

The difference between the first and third rows is about half a degree by 2050 and nearly a degree by 2100, but the difference in consequences is much larger than the temperature numbers suggest. Half a degree is the difference between a heatwave that kills hundreds and one that kills tens of thousands; between a harvest that fails once a decade and one that fails every other year; between a reef that bleaches and recovers and one that dissolves (IPCC, 2018). The reason is that impacts are non-linear: they depend on thresholds, and thresholds are crossed at particular temperatures rather than spread evenly across the range.

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becomes much harder to date and to defend.

The project was nearly cancelled twice. In the 1960s the funding agency cut the budget on the grounds that the work was routine monitoring rather than research, and Keeling had to fight to keep the instruments running; a decade later similar pressures returned. Each time the record was preserved by a small group of scientists who understood that a time series is only valuable if it is unbroken. The episode is instructive for anyone designing a monitoring system today: continuity is a scientific asset that is easy to destroy and impossible to reconstruct.

What the curve did politically was to make the abstract concrete. Before Mauna Loa, the greenhouse effect was a calculation; after it, it was a graph that a parliamentarian could hold up. The same function is served today by the satellite record of sea level and by the global temperature series, and the lesson generalises beyond climate: a long, clean, public measurement is one of the most powerful instruments a society can build, because it converts a contested claim into a shared reference point.

There is a final, quieter lesson in the project's design. Keeling published his data in full, including the seasonal wiggles that his colleagues found distracting, and he insisted that the calibration gases be exchanged between laboratories so that independent groups could compare results. That practice — open data, shared standards, documented methods — became the template for every subsequent global observing system. It is also the reason the curve has survived political attack: a record that anyone can check is much harder to dismiss than a conclusion that must be trusted.

The Keeling Curve is also a record of scientific persistence. Charles David Keeling began measurements at Mauna Loa in 1958 with funding from the International Geophysical Year, and when that funding lapsed he continued with support from the National Science Foundation, insisting on calibration standards that later allowed the record to be compared across decades. The sawtooth pattern he documented — annual rise and fall driven by Northern Hemisphere vegetation — became one of the most recognisable images in science, and the underlying series is now the longest continuous record of atmospheric carbon dioxide anywhere in the world.

The record has a second kind of test, and it arrived by accident. When Mount Pinatubo erupted in June 1991, it injected roughly twenty million tonnes of sulphur dioxide into the stratosphere, where it formed a sulphate aerosol layer and cooled the global surface by about half a degree for two years (IPCC, 2021). The cooling appeared in the same instruments that measure the Keeling curve, and its magnitude matched what the physics predicted. That episode carries more weight than it is usually given: the system responded to a known forcing in a measurable way, which is why the response to a known accumulation of carbon dioxide is not seriously in dispute.

2.12 Future Scenarios and Predictions: The Physical Science

Any statement about the physical future of the climate is conditional, and it is worth being explicit about the condition. The atmosphere responds to the total accumulated carbon dioxide, not to the annual rate at which it is emitted. That single fact determines the shape of every projection in this chapter: two trajectories that emit the same

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cumulative total produce roughly the same peak warming, whether they reach that total quickly or slowly. What differs is the damage along the way, the options available for adaptation, and the risk of crossing thresholds that cannot be reversed on human timescales.

The historical record supplies the baseline. Concentrations were stable at about 280 parts per million for six thousand years before 1750; they reached 315 in 1958, about 370 in 2000 and roughly 424 in 2024, with cumulative emissions since 1850 now close to 2,500 billion tonnes of CO2 (Friedlingstein et al., 2024). Global mean temperature has risen about 1.2 °C over the same period and is currently rising at roughly 0.2 °C per decade (IPCC, 2021). Sea level, which lagged the temperature rise for most of the twentieth century, is now accelerating.

The present position can be stated with unusual precision. For a fifty per cent chance of holding warming below 1.5 °C, the remaining carbon budget is roughly 250 billion tonnes of CO2 — about six years of current emissions (Friedlingstein et al., 2023). For 2 °C the budget is larger but still finite, and the policies in place in 2025 imply warming of about 2.7 °C by 2100 rather than the 1.5 °C target governments have endorsed (IPCC, 2022b; Friedlingstein et al., 2024). Three trajectories bracket the plausible range, and each rests on a different assumption about policy, technology and behaviour.

The first assumes rapid decarbonisation: emissions fall by about half by 2030 and reach net zero around mid-century, with carbon removal compensating for residual sources (Friedlingstein et al., 2023). Warming peaks near 1.8 °C and declines slowly thereafter. The second assumes that current policies continue and are strengthened gradually, producing net zero late in the century; warming reaches about 2.7 °C. The third assumes that decarbonisation stalls and that coal, oil and gas remain dominant; warming reaches roughly 4.4 °C, with the largest increases over land and at high northern latitudes (IPCC, 2021, Table 4.5).

A useful thought experiment clarifies what is already committed. If every emission stopped tomorrow, the atmosphere would stop accumulating CO2 within a few years, but global temperature would stabilise rather than fall, and sea level would continue to rise for centuries as the ocean and the ice sheets finished responding to the warming that has already occurred (IPCC, 2021). The commitment is not a reason for fatalism; it is a reason to distinguish between the warming that cannot now be avoided and the warming that can still be prevented. The first determines how much adaptation is necessary, the second determines whether adaptation remains possible.

Regionally, the three trajectories diverge far more than the global averages suggest. Under the high trajectory, the Mediterranean and the Middle East approach conditions in which summer heat becomes a chronic hazard rather than an episodic one; South Asia faces combinations of heat and humidity that exceed the physiological limits of outdoor work for part of the year; and the Arctic warms by several degrees more than the global mean. Under the low trajectory those outcomes are attenuated but not eliminated. The difference between the scenarios is therefore not a difference in global average temperature; it is a difference in which regions remain habitable in the way their inhabitants currently understand the term.

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